

A multi-modal approach to pool boiling dynamics: Acoustic and vibration data for regime classification and heat flux prediction

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ABSTRACT

This paper presents a multi-scale approach to pool boiling regime classification and heat flux prediction by integrating acoustic, vibration, and imaging techniques. Pool boiling experiments are conducted on a wire heater, and data is collected using microphones, accelerometers, and high-speed cameras. Power spectral density analysis is performed on the acoustic and vibration data, revealing distinct frequency responses that correlate with bubble dynamics and heat flux variations. The spectral power distribution is then used to classify the boiling regimes, identifying transitions from nucleate boiling to critical heat flux (CHF). Proper orthogonal decomposition (POD) and dynamic mode decomposition (DMD) are applied to correlate the boiling frequencies and observe the structural dynamics of the bubbles. POD captured over 98% of the system's energy in the first mode, while DMD identified significant frequency modes between 917.24 Hz and 9799.12 Hz, highlighting key transitions in the boiling process. Simple binary decision tree models are then employed to evaluate the performance of both acoustic and vibration datasets in predicting heat fluxes. The models achieved R^2 scores consistently above 0.99. Principal component analysis (PCA) reduced the complexity of the acoustic and vibration datasets by over 97% while retaining 99% of the variance. The results demonstrate that binary decision tree models are sufficient for predicting heat fluxes with relatively low error. This study provides a framework for real-time classification of boiling regimes and heat flux prediction, offering valuable insights into the dynamics of pool boiling and contributing to the development of safe thermal management systems.

1. Introduction

Boiling heat transfer has emerged as a promising alternative to single-phase systems for high-density cooling applications due to its efficient latent heat transfer mechanism. It is widely used across various applications, from microelectromechanical systems in domestic settings to large-scale power plants in industrial environments. The foundation of boiling heat transfer research is established by Nukiyama [1], who introduced the first boiling curves in the 1970s. These curves depicted the relationship between wall superheat and heat flux derived from pool boiling experiments using wire heaters. Despite these early contributions, a comprehensive understanding of the boiling process remains elusive [2,3]. This gap in knowledge is often attributed to the sensitivity of the phase change process to the thermophysical properties of the heater surface and the boiling fluid. Consequently, most industrial boiling heat transfer systems are designed to operate well below the critical heat flux (CHF) point to ensure stable performance.

CHF denotes the maximum heat transfer rate achievable under nucleate boiling conditions [4]. Accurate prediction of CHF is crucial as it not only enables higher heat transfer rates but also ensures the safe operation of thermal cooling systems. A slight increase in temperature or heat flux beyond the CHF can lead to the formation of a vapor blanket, which drastically reduces heat transfer efficiency and may cause catastrophic damage to the system [5]. Broadly, the existing literature on boiling can be categorized into two main areas: the enhancement of boiling heat transfer [6,7] and the prediction of CHF [8,9]. Enhancing boiling heat transfer generally shifts the CHF to a higher temperature range, allowing the system to operate under safer conditions [10,11]. On the other hand, developing a generalized correlation for CHF prediction is highly challenging due to its dependence on various experimental and environmental variables [12–14]. Unlike conventional methods that primarily rely on steady-state thermodynamic correlations and simplified models of heat transfer,

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Symbol	Description
α	Heat flux ratio, defined as the ratio of heat flux to the critical heat flux (CHF)
A	Dataset matrix representing image intensities
A_1, A_2	Dataset matrices used in Dynamic Mode Decomposition
CHF	Critical heat flux
D_1	Diagonal matrix containing eigenvalues from singular value decomposition
DMD	Dynamic Mode Decomposition
f_i	Angular frequency of the i th mode in DMD
g_i	Growth rate of the i th mode in DMD
I_{iqp}	Intensity value at row q , column p , and time step t
L	Linear mapping matrix between dataset matrices A_1 and A_2
M	DMD mode matrix
$MAPE$	Mean Absolute Percentage Error
MSE	Mean Squared Error
PCA	Principal Component Analysis
POD	Proper Orthogonal Decomposition
PSD	Power Spectral Density
R^2	Coefficient of determination
U, V	Matrices of orthogonal vectors from singular value decomposition
λ_i	Eigenvalue corresponding to the i th mode in DMD
Δt	Time step interval used in DMD analysis

multi-modal approaches [15,16] offer significant advantages by capturing real-time, dynamic interactions between the fluid and surface. Conventional methods often fail to account for transient phenomena such as bubble nucleation, growth, coalescence, and detachment, which are critical to accurately predicting heat flux and CHF. In contrast, multi-modal approaches utilize acoustic, vibration, and imaging techniques to simultaneously monitor pressure wave propagation, mechanical surface responses, and visual bubble dynamics. This integration enables a deeper understanding of boiling transitions and allows for the direct correlation of spectral changes with dynamic events. By capturing these interactions at various scales, multi-modal techniques provide enhanced predictive accuracy, robustness under varying operating conditions, and a framework for real-time monitoring—capabilities that are beyond the reach of traditional methods [17].

The acoustic pressure fluctuations generated during the boiling process can be detected using a microphone, hydrophone, or accelerometer [18]. Of these, the hydrophone is the most sensitive in terms of minimizing interference from noise and addressing impedance mismatch issues at fluid-air interfaces [19]. Bubble formation at temperatures above saturation primarily induces oscillations in the surrounding fluid due to sudden pressure release. These pressure fluctuations occur when bubbles detach from the heater surface, merge with other vapor bubbles, and collapse due to condensation in sub-cooled conditions. The characteristic frequencies associated with bubble generation, oscillation, and merging in a chaotic boiling environment can be detected by sensors via the corresponding acoustic pressure fluctuations [20]. As these oscillations propagate through the cloud of bubbles in the liquid medium, they may occasionally modulate into other frequencies with disturbed amplitudes [21].

Moreover, the frequencies and corresponding amplitudes detected by the sensors are not the original values but are closely linked to the underlying bubble dynamics. While several researchers have successfully identified spectral variations at different heat flux levels throughout the boiling curve, none have established a physical mechanism that directly correlates these spectral variations with the structural dynamics of boiling bubbles [16,17,22]. This study addresses this critical gap by taking a step forward in establishing a relationship between spectral

observations and bubble dynamics in pool boiling, using a wire heater as the test case and employing a multi-modal approach.

This study explores the multi-scale structural dynamics underlying the characteristic frequencies observed in the spectral plots of vibrations and acoustic measurements during pool boiling. These measurements are recorded simultaneously and analyzed in conjunction with proper orthogonal decomposition (POD) [23–25] and dynamic mode decomposition (DMD) [26], both derived from high-speed images captured during the boiling process. The study also utilizes principal component analysis (PCA) [27] to reduce the complexity of the datasets while retaining essential information. Although the high-speed images are not recorded at the exact same time as the acoustic and vibration measurements, they are taken under the same steady-state conditions with minimal time differences, ensuring comparability of the frequency data. The study further evaluates the effectiveness of binary decision tree models [28] for heat load prediction, showing that these models achieve high accuracy despite the varying complexity of the boiling regimes.

The following sections of the paper are arranged as follows: Section 2 details the materials and methods used in the investigation, including a brief theory on proper orthogonal decomposition and dynamic mode decomposition. Section 3 describes the results and discussion, and Section 4 details the conclusions from the investigations with limitations.

2. Materials and methods

The experimental setup (as shown in Fig. 1) [16] employed in this study consists of a transparent glass boiling chamber with the dimensions of 12 inches in length, 9 inches in width, and 12 inches in height. This chamber features a removable top cover to facilitate access to the internal components during the experiments. Within the chamber, two copper electrodes are installed. These electrodes are equipped with a screw-type base that allows for the secure mounting and adjustment of the heating elements essential for the experiments. The electrodes support a nichrome wire of SWG 40 gauge which is used as the boiling surface for observing bubble dynamics in a saturated boiling condition.

Temperature control within the chamber is achieved through the use of four cartridge heaters of 250 watts each. These heaters are connected to a temperature relay system, which is calibrated to ensure that the water within the chamber is heated to and maintained at the saturation temperature necessary for consistent experimental conditions. The temperature of the chamber is monitored by four Resistance Temperature Detectors (RTDs). These RTDs are connected to a “USB-Temp” device which is an 8-channel temperature measurement instrument capable of sampling at a 2 Hz frequency.

Acoustic and vibration data collection is a critical component of the experimental setup. A $\frac{1}{4}$ -inch free-field microphone (377A01 /Briel & Kjaer) is strategically placed within the chamber to capture the acoustic emissions produced during the boiling process. Additionally, an accelerometer (4513/Briel & Kjaer) is installed to monitor vibrations induced by the boiling. Both sensors are integrated into a comprehensive data acquisition system that includes a PCB signal conditioner, a low-pass filter, and an 8-channel simultaneous sampling board. This setup pre-processes and amplifies the signals from the sensors. The experimental activities are conducted within a semi-anechoic chamber situated at the Indian Institute of Technology Madras, India. To ensure that the experimental environment is controlled and isolated, the boiling chamber is lined with polyurethane foam which acts as an insulation against external vibrations. The experimental setup has an estimated heat flux uncertainty of approximately $\pm 2.48\%$, primarily due to voltage and current measurement tolerances. Sensor calibration uncertainties are ± 1 dB for the microphone in the 4 Hz to 7 kHz range, as detailed in our previous work [16], and $\pm 10\%$ for the accelerometer up to 10 kHz.

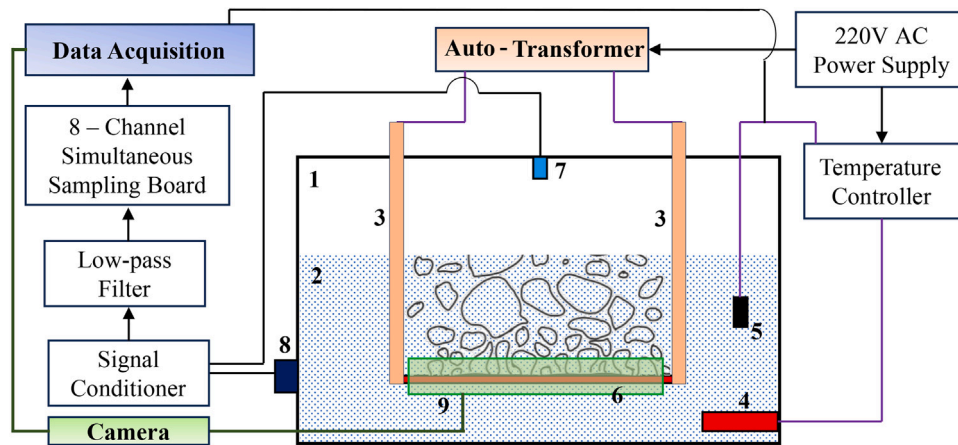


Fig. 1. Schematic view of the experimental setup detailing data acquisition. 1 - Boiling Chamber, 2 - Water medium, 3 - Electrodes, 4 - Cartridge heater, 5 - Thermocouple, 6 - Heater element, 7 - Microphone, 8 - Accelerometer, 9 - Camera view window. Please note that the image is not drawn to scale.

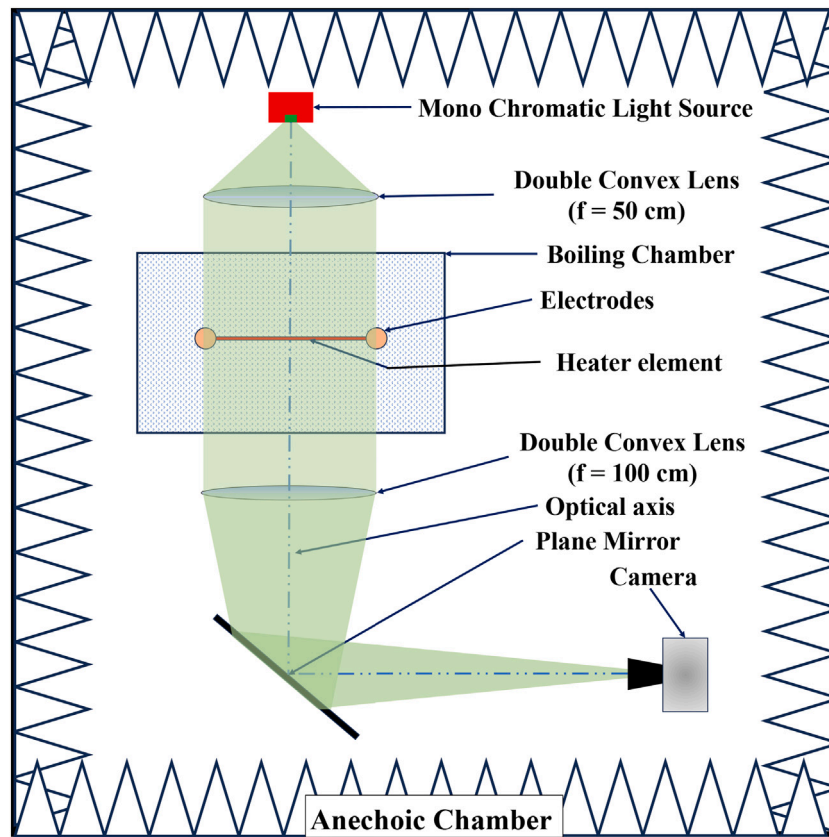


Fig. 2. Schematic top-view of the experimental setup showing optical axis, placement of light source, lens, mirror, and camera within the anechoic chamber with a cutoff frequency of 700 Hz. 'f' denotes the focal length of the lens. Please note that the image is not drawn to scale.

Shadowgraphy technique (as shown in Fig. 2) is utilized for the visual analysis. This technique involves a monochromatic direct current-based LED green light source that emits light which is transformed into a parallel beam by a double-convex lens with a focal length of 50 cm. This beam passes through the boiling chamber, and a subsequent double-convex lens with a 100 cm focal length focuses the light onto a high-speed camera. A Photron Mini UX 100 camera, which is capable of capturing up to 40,000 frames per second, is used.

The water bath temperature is controlled by the cartridge heaters connected to the temperature relay system, which is initially set to a target temperature of 98 °C. This system gradually increases the water temperature to the saturation point. After reaching the near saturation

temperature, the electrical input to the heater elements begins and advances to various manually selective voltages to increase the heat flux through an auto-transformer. Voltage increments are monitored, and acoustic and vibration data is captured at each steady stage at a sampling frequency of 150 kHz. This monitoring captures the evolution of acoustic, vibration, and temperature profiles throughout the boiling process until CHF.

2.1. Theory of decomposition

In this section, the theory of decomposition methods is discussed with a focus on POD and DMD, which are considered the two most

Table 1
Pixel array representation in image data.

I'_{11}	I'_{12}	I'_{13}	...	...	...	I'_{1p}
I'_{21}	I'_{22}	I'_{23}	...	...	...	I'_{2p}
I'_{31}	I'_{32}	I'_{33}	...	...	...	I'_{3p}
...	...	...	...	...	...	...
...	...	...	...	...	...	...
I'_{q1}	I'_{q2}	I'_{q3}	...	...	...	I'_{qp}

powerful tools for analyzing complex fluid dynamics and thermal systems. POD is a technique that captures the most energetic modes of a system by identifying dominant patterns from a set of spatial data. It provides a low-dimensional representation that retains the key features of the original data. On the other hand, DMD extends this analysis by uncovering dynamic modes associated with specific frequencies, capturing both spatial and temporal evolution within the system. Together, POD and DMD offer complementary approaches to understanding the underlying structures and dynamics in complex systems.

2.1.1. Proper orthogonal decomposition (POD)

POD [29] stands as a potent technique for constructing reduced order models (ROM) from complex, high dimensional datasets. It distills complex data into a lower-dimensional representation that encapsulates its inherent characteristics. This is achieved through an orthogonal transformation of numerous variables within the dataset generating a set of uncorrelated linear principal components. The number of these principal components is often fewer than the original dataset's dimensions. In the context of the ongoing study, the dataset encompasses pixel intensity information extracted from high-speed images. Employing POD analysis on these images affords a glimpse into the temporal dynamics of bubbles and heater elements. The image data contains pixel information arranged in the form of an array as shown in Table 1.

Each element in the Table 1 represents the intensity information, and I'_{qp} refers to the intensity value at q th row, p th column at t th time step. This data of each image is arranged into each column of the data set matrix $\mathbf{A}$. Hence, the data set matrix contains all the images information taken at periodic intervals of time. The matrix $\mathbf{A}$ is given as

$$\mathbf{A} = \begin{bmatrix} I_{11}^1 & I_{11}^2 & I_{11}^3 & \dots & \dots & I_{11}^t \\ I_{12}^1 & I_{12}^2 & I_{12}^3 & \dots & \dots & I_{12}^t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{1p}^1 & I_{1p}^2 & I_{1p}^3 & \dots & \dots & I_{1p}^t \\ I_{21}^1 & I_{21}^2 & I_{21}^3 & \dots & \dots & I_{21}^t \\ I_{22}^1 & I_{22}^2 & I_{22}^3 & \dots & \dots & I_{22}^t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{2p}^1 & I_{2p}^2 & I_{2p}^3 & \dots & \dots & I_{2p}^t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{q1}^1 & I_{q1}^2 & I_{q1}^3 & \dots & \dots & I_{q1}^t \\ I_{q2}^1 & I_{q2}^2 & I_{q2}^3 & \dots & \dots & I_{q2}^t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{qp}^1 & I_{qp}^2 & I_{qp}^3 & \dots & \dots & I_{qp}^t \end{bmatrix}$$

The singular value decomposition of this matrix $\mathbf{A}$ gives the factorization of $\mathbf{A}$ into three different matrices, i.e., $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T$, where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal vectors. The matrix $\mathbf{D}$ is the diagonal matrix with

the diagonal elements as eigen values. The columns of $\mathbf{U}$ contain the image information in the order of highest to lowest energy content, and columns of $\mathbf{V}$ contain the time series data of the corresponding image intensity fluctuations. The frequency spectra from the columns of $\mathbf{V}$ give the corresponding modes of columns $\mathbf{U}$.

2.1.2. Dynamic mode decomposition (DMD)

DMD [26] is a relatively new technique in the reduced order models research area. Unlike POD, which gives more importance to the energy content in the data set, DMD considers the spatio-temporal patterns. Besides, DMD constructs the reduced order models from the non-orthogonal datasets. Hence, DMD is based on the frequency content rather than the energy content. This leads the DMD modes to associate with a distinctive frequency.

For the DMD analysis, two data-sets need to be created from the main data-set to induce a single frame lag between the data-sets. The two data-sets $\mathbf{A}_1$ and $\mathbf{A}_2$ are created from the main data-set $\mathbf{A}$ as follows

$$\mathbf{A}_1 = \begin{bmatrix} I_{11}^1 & I_{11}^2 & I_{11}^3 & \dots & \dots & I_{11}^{t-1} \\ I_{12}^1 & I_{12}^2 & I_{12}^3 & \dots & \dots & I_{12}^{t-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{1p}^1 & I_{1p}^2 & I_{1p}^3 & \dots & \dots & I_{1p}^{t-1} \\ I_{21}^1 & I_{21}^2 & I_{21}^3 & \dots & \dots & I_{21}^{t-1} \\ I_{22}^1 & I_{22}^2 & I_{22}^3 & \dots & \dots & I_{22}^{t-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{2p}^1 & I_{2p}^2 & I_{2p}^3 & \dots & \dots & I_{2p}^{t-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{q1}^1 & I_{q1}^2 & I_{q1}^3 & \dots & \dots & I_{q1}^{t-1} \\ I_{q2}^1 & I_{q2}^2 & I_{q2}^3 & \dots & \dots & I_{q2}^{t-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{qp}^1 & I_{qp}^2 & I_{qp}^3 & \dots & \dots & I_{qp}^{t-1} \end{bmatrix}$$

$$\mathbf{A}_2 = \begin{bmatrix} I_{11}^2 & I_{11}^3 & I_{11}^4 & \dots & \dots & I_{11}^t \\ I_{12}^2 & I_{12}^3 & I_{12}^4 & \dots & \dots & I_{12}^t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{1p}^2 & I_{1p}^3 & I_{1p}^4 & \dots & \dots & I_{1p}^t \\ I_{21}^2 & I_{21}^3 & I_{21}^4 & \dots & \dots & I_{21}^t \\ I_{22}^2 & I_{22}^3 & I_{22}^4 & \dots & \dots & I_{22}^t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{2p}^2 & I_{2p}^3 & I_{2p}^4 & \dots & \dots & I_{2p}^t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{q1}^2 & I_{q1}^3 & I_{q1}^4 & \dots & \dots & I_{q1}^t \\ I_{q2}^2 & I_{q2}^3 & I_{q2}^4 & \dots & \dots & I_{q2}^t \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{qp}^2 & I_{qp}^3 & I_{qp}^4 & \dots & \dots & I_{qp}^t \end{bmatrix}$$

These $\mathbf{A}_1$ and $\mathbf{A}_2$ are linearly mapped with another matrix $\mathbf{L}$ as $\mathbf{A}_2 = \mathbf{L}\mathbf{A}_1$. As the data set matrices are not square matrices, the

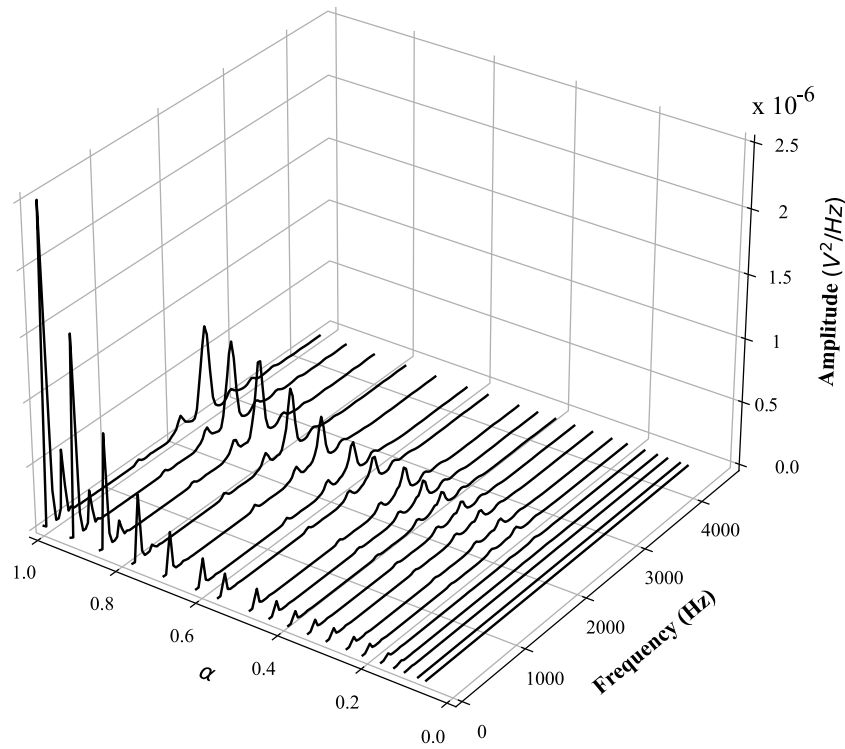


Fig. 3. Acoustic spectra.

direct solution of $\mathbf{L}$ is highly difficult and computationally expensive. Hence, initially, an economy sized singular value decomposition of $\mathbf{A}_1$ is performed. This leads to $\mathbf{A}_1 = \mathbf{U}_1 \cdot \mathbf{D}_1 \cdot \mathbf{V}_1^T$. As mentioned in the POD analysis, the matrix $\mathbf{U}_1$ has the POD modes, $\mathbf{D}_1$ contains the eigen values representing the energy content in the data, and $\mathbf{V}_1$ contains the temporal evolution corresponding POD modes across the span. If a reduced order model is considered, then a reduced order matrix $\hat{\mathbf{L}}$ is to be constructed from $\mathbf{L} \approx \hat{\mathbf{L}} = \mathbf{U}_1^T \cdot \mathbf{A}_2 \cdot \mathbf{V}_1 \cdot \mathbf{D}_1^{-1}$. The eigen-decomposition of matrix $\hat{\mathbf{L}}$ gives eigen vector matrix $\mathbf{W}$ and eigen value diagonal matrix λ . Accordingly, DMD modes can be acquired by $\mathbf{M} = \mathbf{A}_2 \cdot \mathbf{V}_1 \cdot \mathbf{D}_1^{-1} \cdot \mathbf{W}$ and the corresponding angular frequency of i th mode is obtained by $\mathbf{f}_i = [\text{Imaginary}(\log \lambda_i)/\Delta t]$, the growth rate is given by $\mathbf{g}_i = [\text{Real}(\log \lambda_i)/\Delta t]$.

3. Results and discussion

In this section, the results from the multi-modal investigation into pool boiling regime classification are presented and discussed. The findings are organized into five key areas. First, the comparison of power spectral density highlights the distinct frequency responses across different boiling regimes. Second, the identification of CHF is achieved through spectral power distribution analysis. Third, proper orthogonal decomposition (POD) is employed to extract dominant flow features, followed by DMD to capture the system's dynamic modes. Finally, the performance of various machine learning methods is evaluated using acoustic and vibration data-sets, comparing their effectiveness in regime classification.

3.1. Comparison of acoustic and vibration power spectral density for various heat fluxes along pool boiling curve

The time series data collected through simultaneous microphone and accelerometer recordings are processed to generate power spectral density (PSD) plots in the frequency domain. Figs. 3 and 4 display the PSD graphs derived from the microphone and accelerometer data. The α in the plots represents the ratio of heat flux to the CHF. Hence, $\alpha = 1$

represents the CHF condition. In the acoustic spectra, peak amplitudes increase with heat flux, indicating more energy released as sound with rising heat flux. This trend reflects boiling activities such as bubble generation, merging, and collapse, influenced by the surrounding medium's impedance. Conversely, in the vibration spectra, peak amplitudes rise until a certain threshold before declining, suggesting that increases in heat flux initially enhance bubble formation and growth but eventually dampen surface vibrations as the CHF is approached.

The analysis of both acoustic and vibration spectra provides a comprehensive understanding of the boiling process. The increasing amplitudes in the acoustic data correlate with intensifying bubble activity, while the trends in the vibration data illuminate the mechanical impacts of bubble interactions on the boiling surface. These observations are crucial for identifying CHF as they are marked by significant changes in heat transfer and boiling dynamics.

Further analysis reveals that acoustic frequency shifts are less pronounced than previously reported [16] in studies involving thin-diameter wires. The accelerometer spectral plots (Fig. 6) show a peak amplitude at 9 kHz, which increases until 48% of the CHF and then decreases. At CHF, the amplitude returns to levels observed at the onset of boiling, with spectral power shifting toward lower frequencies. Beyond 50% of CHF, amplitudes at lower frequencies increase, indicating larger mean bubble diameters at higher heat fluxes. These findings refine the present understanding of the boiling process. Unlike earlier research that highlighted significant acoustic frequency shifts in thin-diameter wires, present data suggest these shifts are subtler. Instead, critical insights emerge from the accelerometer data, particularly the behavior of the 9 kHz peak amplitude. This peak's initial rise correlates with increasing heat flux up to nearly half of CHF, reflecting intensified bubble activity and surface vibrations.

As CHF nears, the decline in the 9 kHz amplitude to levels seen at boiling inception marks a pivotal transition in boiling behavior, signaling the shift from nucleate boiling to the CHF regime (Fig. 5). The accompanying shift of spectral power to lower frequencies at CHF suggests changes in bubble dynamics, specifically larger bubbles forming as heat flux continues to increase. The dominance of lower

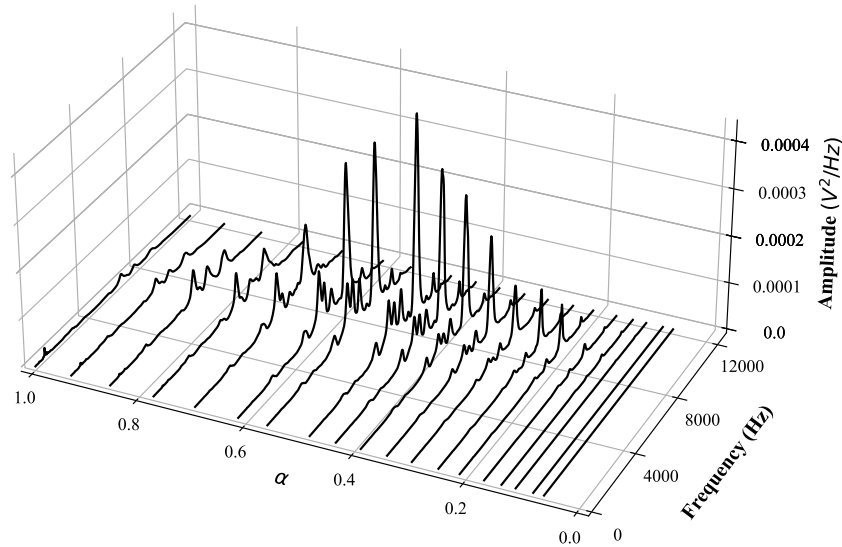


Fig. 4. Vibration spectra.

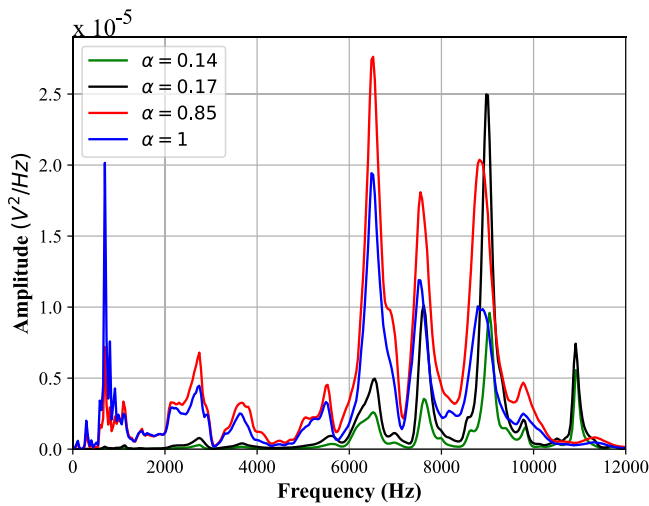


Fig. 5. Vibration spectral amplitude variation of near boiling incipience and CHF.

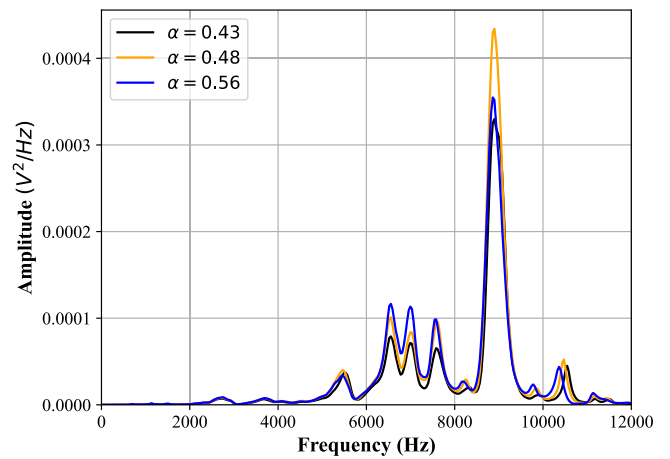


Fig. 6. Vibration spectral amplitude variation of 9 kHz near 50% of CHF.

frequencies and the increased power at these frequencies imply that larger bubbles are increasingly influential in the boiling process at higher heat fluxes. This trend suggests that as bubbles grow larger, their formation and detachment frequencies decrease, leading to a more pronounced presence of lower frequency components in the vibration spectra. This insight is essential for understanding the transition phases in boiling regimes, especially around CHF, where there is a significant transformation in heat transfer efficiency.

The spectra near the boiling incipience are meticulously analyzed to evaluate the sensitivity of each measurement method. Figs. 7 and 8 showcase these analyses, revealing the accelerometer's heightened sensitivity to boiling incipience compared to the microphone. This enhanced sensitivity is due to the accelerometer's ability to detect subtle surface vibrations caused by the nucleation, growth, and collapse of small bubbles on the heating surface. These vibrations produce distinct and well-defined peaks in the accelerometer spectra, even at early heat flux levels, making the vibration data highly reliable for detecting the onset of boiling.

Conversely, while the acoustic spectra also exhibit small peaks at incipience, these peaks are weak and comparable in magnitude to the curved baseline observed when no boiling occurs. This makes the

acoustic signals less distinguishable from background noise, limiting their effectiveness in robust detection scenarios. Although acoustic signals can be detected in controlled laboratory environments, their sensitivity to external noise and interference poses challenges for practical applications.

The comparison of high and low-frequency spectral powers further illustrates the progression of boiling. At boiling incipience, high-frequency components dominate, reflecting the rapid formation and collapse of small bubbles. As heat flux increases and larger bubbles form, low-frequency components become more prominent, signifying the transition toward the critical heat flux (CHF). The vibration spectra consistently highlight these transitions, providing a more robust framework for identifying key boiling stages compared to the acoustic data.

In terms of CHF detection, analyzing shifts from high to low-frequency spectral powers can yield important insights [17,30]. As heat flux nears CHF, the decline in high-frequency power coupled with an increase in low-frequency power can signal this transition. The accelerometer's superior sensitivity to surface vibrations makes it especially valuable in capturing these spectral shifts, enhancing its utility for early CHF detection [20]. The subsequent analysis will delve deeper into these findings, exploring how detailed spectral analysis can refine CHF detection techniques. By comprehending the specific

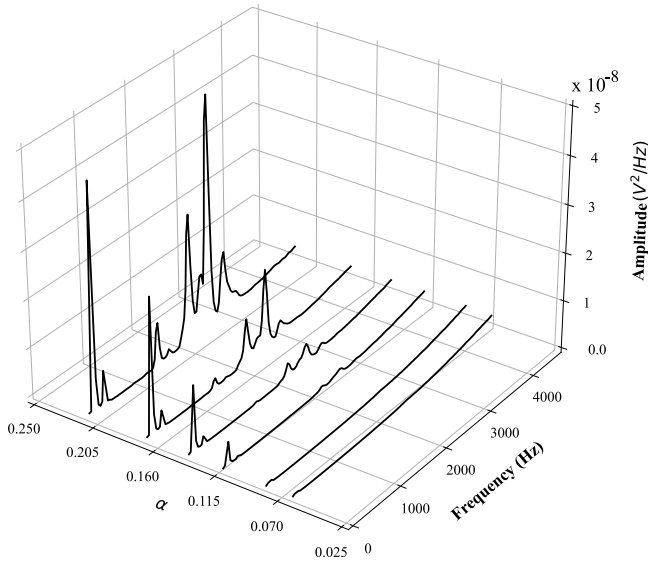


Fig. 7. Acoustic spectra at boiling incipience.

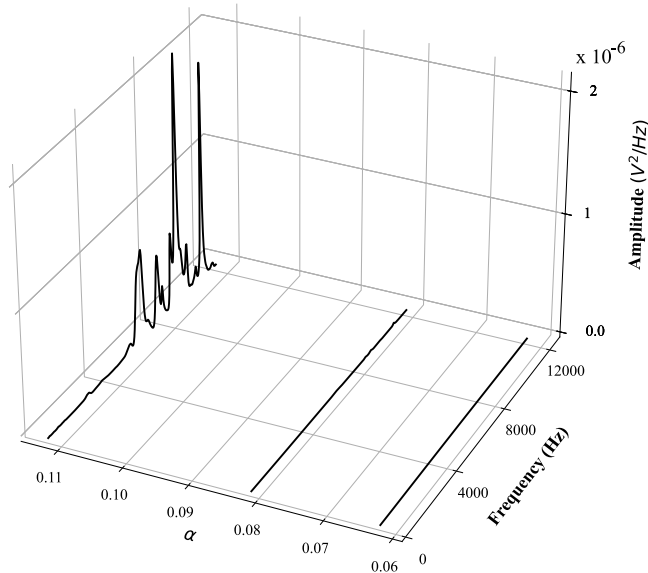


Fig. 8. Vibration spectra at boiling incipience.

spectral characteristics associated with different boiling regimes and transitions, more accurate and reliable methods for predicting and identifying CHF can be developed. This advancement can significantly improve the safety and efficiency of thermal management systems, ensuring optimal performance across various operational conditions.

3.2. CHF identification based on spectral power distribution

In the previous section, it is established that the spectral power distribution is closely aligned with the dynamics of boiling, highlighting its importance in understanding boiling phenomena. This section delves deeper into the distribution of spectral power, focusing on its relevance to identifying CHF and providing a dual analysis of both vibration and acoustic spectra to yield comprehensive insights.

For this study, cumulative power spectral density is calculated across specifically chosen frequency ranges to encapsulate the distribution of spectral power at both low and high frequencies. In the acoustic

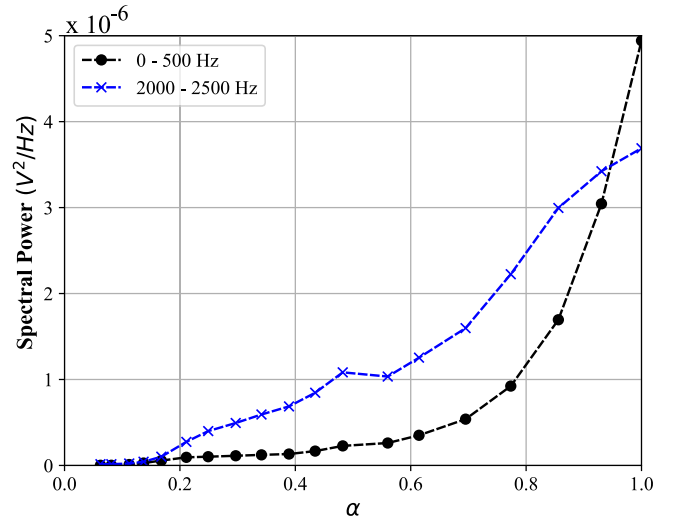


Fig. 9. Cumulative acoustic spectral power amplitudes at specific intervals.

analysis, the low-frequency range is defined as 0–500 Hz, and the high-frequency range is set between 2000 and 2500 Hz to include all the characteristic peaks observed in the spectra. For the vibration spectral power analysis, the low-frequency range is established between 6000 and 8000 Hz to highlight vibrations linked to bubble generation and behavior, while the high-frequency range is set from 8000 to 10000 Hz. These frequency intervals are carefully chosen to ensure comprehensive coverage of the key dynamic peaks within each spectrum. The interval lengths are kept constant across both the acoustic and vibration spectra analyses, which are 500 Hz for acoustic data and 2000 Hz for vibration data.

Figs. 9 and 10 illustrate the cumulative spectral power distributions across the selected frequency ranges for both acoustic and vibration spectra. The acoustic spectral power distribution graph shows that while the high-frequency interval predominates at most heat flux levels, it notably peaks near 70% of the heat flux, aligning with previous observations [17,18]. At CHF, however, low-frequency amplitudes become more prominent [17]. In contrast, the vibration spectral power distribution demonstrates that lower frequency intervals tend to dominate at CHF, though the differences between the spectral powers are less marked than in the acoustic data.

The thorough examination of spectral power distributions not only underscores the utility of both acoustic and vibration data in identifying CHF but also shows how an increase in high-frequency spectral power can signal the approach to CHF. Although the vibration data provides a subtler contrast, it effectively complements the acoustic data, offering additional verification. Together, these findings enhance the understanding of boiling dynamics and contribute to developing more precise and reliable methods for detecting and managing CHF in thermal systems.

To solidify these findings, Figs. 11 and 12 illustrate the slope of spectral power variations across different heat fluxes. The slope graph for acoustic spectral power shows a clear trend, with low-frequency amplitudes surpassing high-frequency amplitudes at CHF. This clear distinction in slopes highlights the sensitivity of acoustic data to the changes occurring near CHF. On the other hand, the spectral power slope for vibration spectra is less conclusive due to similar slope trends for both intervals. Despite this, the declining amplitudes in vibration spectra with increasing heat flux result in negative spectral power slopes for both intervals. This negative trend in spectral power slopes suggests that while vibration data may not distinctly separate low and high-frequency intervals near CHF, it still provides useful information. Specifically, vibration spectra are particularly effective for detecting

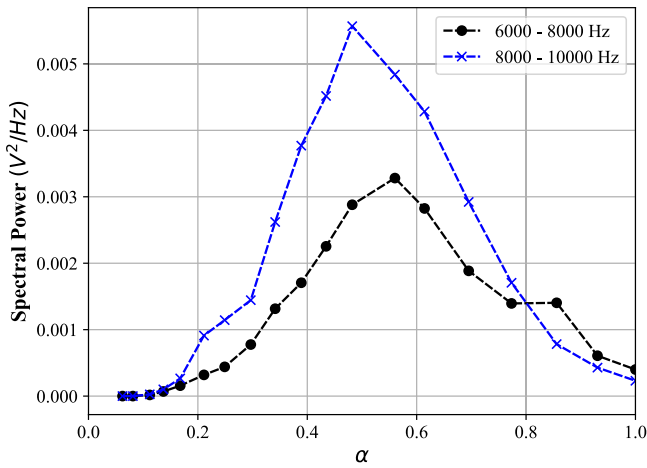


Fig. 10. Cumulative vibration spectral power amplitudes at specific intervals.

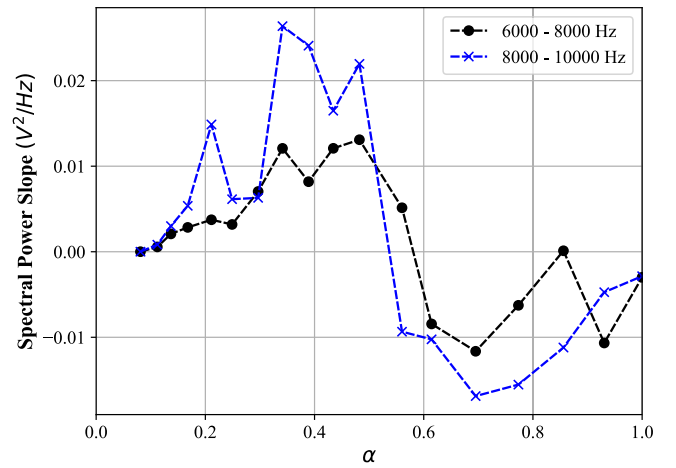


Fig. 12. Cumulative vibration spectral power amplitude slope at specific intervals.

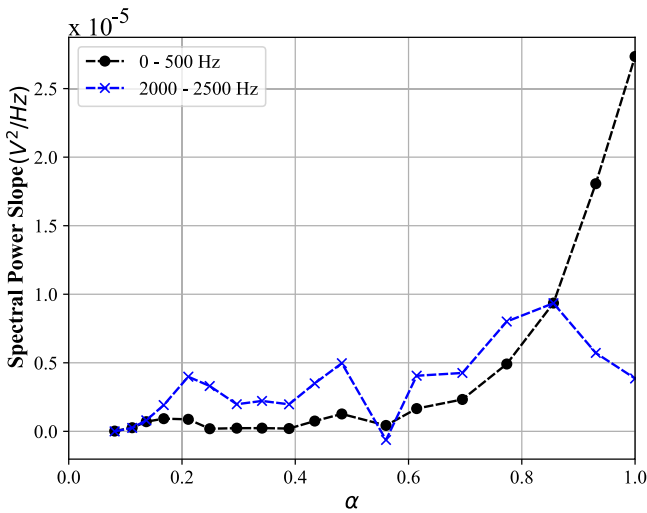


Fig. 11. Cumulative acoustic spectral power amplitude slope at specific intervals.

boiling incipience due to their sensitivity to the initial formation and growth of bubbles on the boiling surface.

Furthermore, the rise and fall of peak amplitudes around 50% of CHF in the vibration spectra can also assist in identifying CHF. This characteristic behavior, where peak amplitudes increase up to a certain point and then decline as CHF is approached, provides an additional marker for detecting this critical transition. Overall, the analysis of spectral power slopes reinforces the complementary nature of acoustic and vibration data in identifying CHF. While acoustic spectra offer a more pronounced indication of CHF through distinct slope variations, vibration spectra provide early detection capabilities and additional confirmation through amplitude trends.

3.3. Mean image visualization of pool boiling dynamics

Before diving into the details of POD and its modes, it is crucial to present the mean images of boiling at various heat fluxes. These mean images serve as a baseline for understanding the boiling phenomena, offering a clear view of the average spatial patterns across different conditions. By examining these visual baselines, the subsequent analysis of the POD modes will be more insightful, helping to distinguish significant features and deviations.

Figs. 13 and 14 show the mean images of the boiling phenomenon on the wire heater at various heat fluxes. Each mean image represents

the average boiling behavior over time, with the heat flux ratio (α) ranging from 0.06 to 1.0. The heat flux ratio is defined as the ratio of heat flux to the CHF. At low heat flux ratios ($0.06 \leq \alpha < 0.21$), bubble formation is minimal, and the boiling surface remains largely undisturbed. The dominant heat transfer mode is natural convection, with only a few small bubbles sparsely distributed across the surface. As the heat flux ratio increases ($0.21 \leq \alpha < 0.43$), the boiling process transitions to the nucleate boiling regime. Bubble nucleation intensifies, leading to visible streaks on the heater surface as bubbles generate and coalesce, enhancing heat transfer.

In the fully developed nucleate boiling range ($0.43 \leq \alpha < 0.69$), vigorous bubble activity and frequent coalescence create distinctive patterns on the heater surface. The boiling surface is fully active, with bubbles departing regularly and leaving streaks that illustrate the vigorous heat transfer process. As the heat flux ratio continues to increase ($0.69 \leq \alpha < 0.93$), the boiling process approaches the CHF. The boiling surface shows mixed characteristics of nucleate and transition boiling, resulting in irregular streaks and patches. Increased bubble coalescence and partial vapor film formation create instability in the boiling process.

At the CHF ($0.93 \leq \alpha < 1.0$), the boiling surface experiences high levels of instability. The heat flux reaches its peak, and the boiling process becomes highly unstable. The mean images reveal a chaotic pattern, with fewer distinct bubbles and a higher prevalence of vapor patches, indicating partial dry-out and impending transition to film boiling. By comparing the mean images to the POD modes, it becomes easier to recognize significant features, anomalies, and deviations that inform the understanding of boiling phenomena at various stages, particularly as the system approaches CHF.

3.4. Results of proper orthogonal decomposition

The accelerometer spectra, POD analysis, and microphone spectra offer a comprehensive view of boiling-induced phenomena, each revealing unique vibrational and acoustic characteristics. The accelerometer spectra as shown in Fig. 15 depict the distribution of vibrational energy across a wide frequency range. Significant energy is concentrated in the low-frequency band, with prominent peaks below 1000 Hz, indicating strong boiling-induced vibrations. Additionally, there are notable peaks at approximately 6000 Hz and 8000 Hz, highlighting the presence of higher frequency vibrations, which signifies the complex vibrational behavior during the boiling process.

POD analysis as shown in Fig. 16 provides insights into the dominant spatial structures within the boiling phenomena. The primary spatial feature is a consistent band at the center of the boiling region.

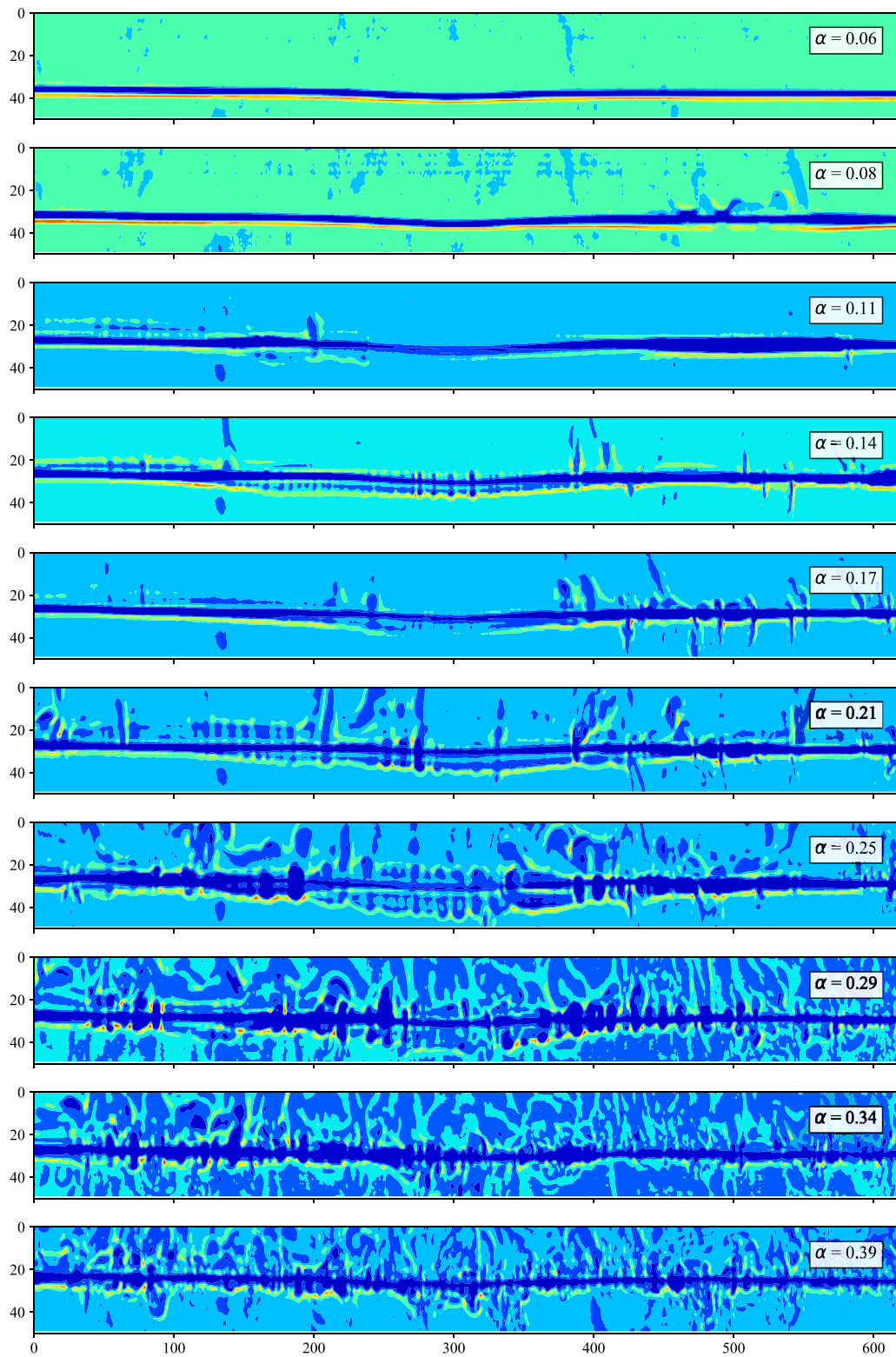


Fig. 13. Mean image at various heat fluxes ($\alpha = 0.06$ to 0.39).

The corresponding POD spectra reveal the frequencies associated with these spatial patterns, showing the highest amplitude below 1000 Hz. Although there are peaks at 6000 Hz and 8000 Hz, they are less pronounced compared to the accelerometer spectra, indicating their reduced impact on the overall spatial structure. The microphone spectra capture the distribution of acoustic energy, offering complementary insights. There are significant peaks below 1000 Hz and around 1500 Hz.

Additionally, there are peaks at approximately 3000 Hz. Frequencies above 4000 Hz exhibit negligible amplitude, suggesting that high-frequency acoustic vibrations are less prominent in boiling-induced phenomena.

A comparative analysis reveals that the accelerometer spectra showcase a broad range of vibrational frequencies up to 10,000 Hz, indicating complex vibrational dynamics during boiling. In contrast, the

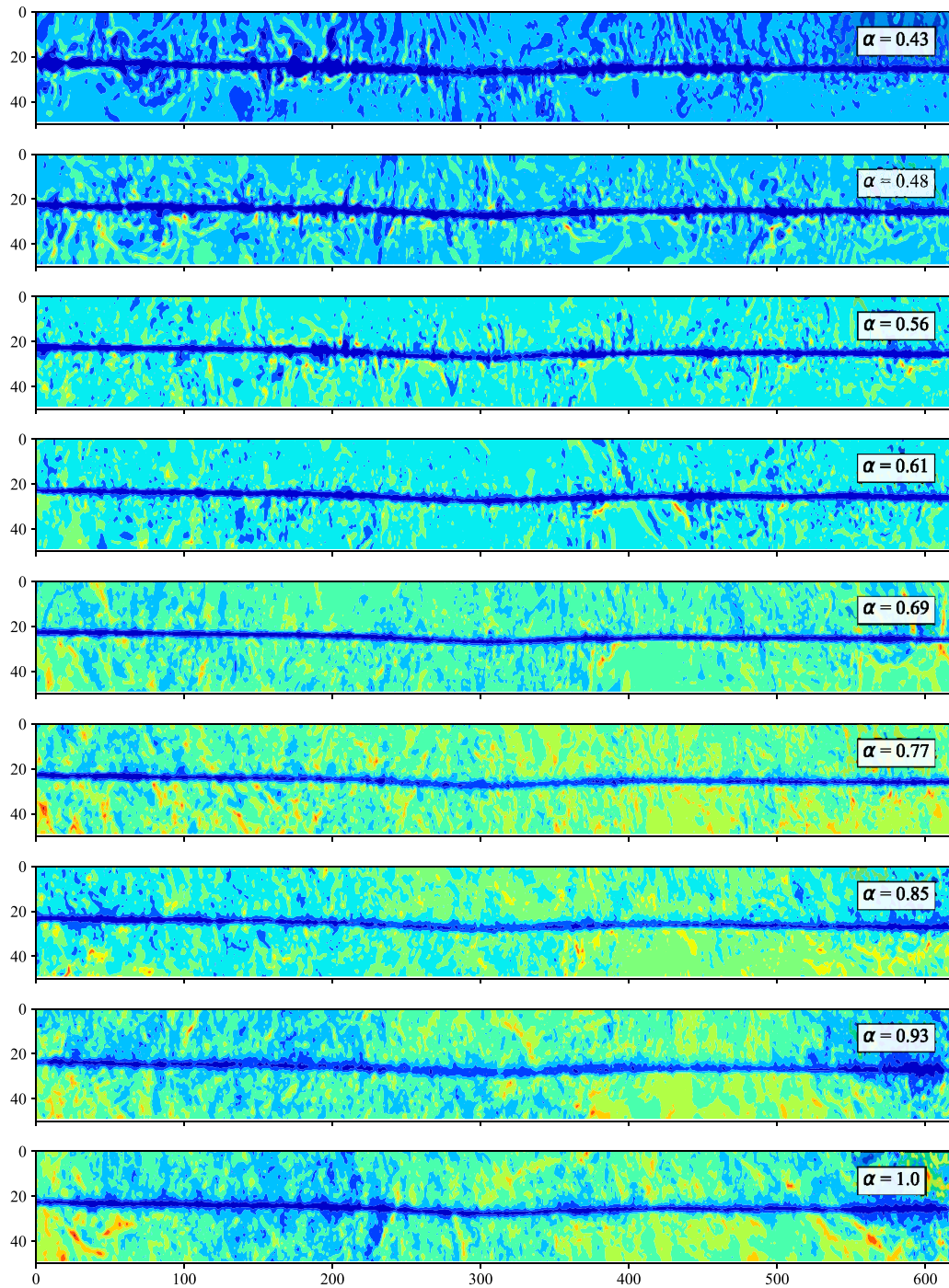


Fig. 14. Mean image at various heat fluxes ($\alpha = 0.43$ to 1).

microphone spectra are dominated by low frequencies below 500 Hz, with additional peaks around 3000 Hz. The POD analysis serves as a bridging tool, highlighting that the primary spatial structures in boiling dynamics are driven predominantly by low-frequency vibrations below 1000 Hz. Higher-frequency vibrations, although detected by the accelerometer, have a lesser role in shaping the overall boiling dynamics.

In summary, the accelerometer spectra reveal a wide range of boiling-induced vibrational frequencies, with significant peaks at lower frequencies and additional peaks at 6000 Hz and 8000 Hz. The microphone spectra emphasize low-frequency acoustic emissions, with notable peaks below 500 Hz and around 3000 Hz. The POD analysis

demonstrates that the primary spatial structures in boiling are influenced mainly by low-frequency vibrations. Higher-frequency peaks play a secondary role in the overall spatial dynamics. The first POD mode is retained in the spectral analysis due to the substantial drop in eigenvalues in subsequent modes, indicating that the primary spatial structure, which is also the mean amplitude, captures the most significant features of the boiling phenomena.

3.5. Results of dynamic mode decomposition

Building upon the spatial insights obtained through Principal Component Analysis (PCA) and Proper Orthogonal Decomposition (POD), Dynamic Mode Decomposition (DMD) further elucidates the temporal

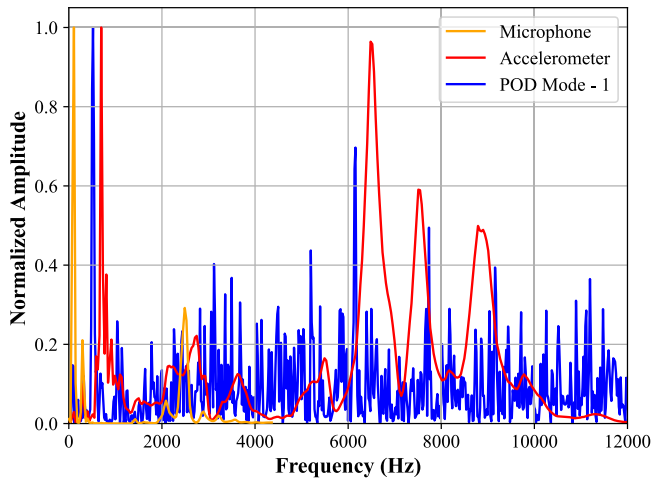


Fig. 15. Comparison of normalized spectral amplitude of microphone, accelerometer, and POD data.

dynamics of boiling by linking spatial structures to specific frequencies. While PCA and POD primarily captured static or averaged features of the boiling process, DMD adds another layer of depth by uncovering how key frequency modes evolve dynamically over time. This analysis enables a more precise understanding of bubble formation, growth, and detachment and how they affect heat flux variations.

In DMD analysis (as shown in Fig. 17) of pool boiling on a wire heater, a frequency range from 917.24 Hz to 9799.12 Hz is examined. Lower frequencies within this range typically correspond to larger, slower-evolving phenomena, such as the growth and coalescence of large bubbles or oscillations in the liquid layer adjacent to the heating surface. For example, frequencies around 917.24 Hz primarily capture large-scale bubble formations and sustained vapor columns, consistent with observations in the POD analysis, where dominant spatial modes were linked to low-frequency vibrational energy. On the other hand, higher frequencies (e.g., 9534.78 Hz) represent smaller, faster-evolving dynamics, including the rapid formation, detachment, and collapse of smaller bubbles.

The presence of peak frequencies in the higher range highlights regions of intense activity, which correspond to critical heat transfer

points where bubble detachment is frequent. These peak frequencies, particularly at 9534.78 Hz, align with the enhanced heat flux conditions identified in the power spectral density analysis and reflect significant bubble interactions. Such correlations suggest that higher-frequency modes detected through DMD are directly linked to heat transfer intensification, especially during the transition from nucleate boiling to critical heat flux (CHF).

Furthermore, DMD analysis reveals that bubble activity is not uniformly distributed along the wire heater. Variations in amplitude across different frequencies indicate spatially localized differences in boiling efficiency. Sections of the wire that exhibit higher amplitudes at critical frequencies correspond to regions of more effective bubble generation and detachment.

By integrating DMD with previous analyses, it becomes evident that the lower-frequency modes identified in both the POD and DMD analyses contribute to the formation of larger bubbles that dominate the boiling surface as the system approaches CHF. These larger bubbles are reflected in the shift of spectral power toward lower frequencies in both the acoustic and vibration datasets. Meanwhile, the higher-frequency modes observed through DMD primarily govern localized, high-intensity events that enhance instantaneous heat transfer. The combined understanding of these low- and high-frequency contributions establishes a clearer picture of how boiling transitions impact overall heat flux and boiling regime classification.

3.6. Performance comparison of machine learning models with acoustic and vibration data-sets

The evaluation of machine learning models provides insights into their effectiveness in predicting boiling behaviors using acoustic and vibration data. Acoustic and vibration signals differ in frequency characteristics and complexity, requiring specific analysis to determine which dataset yields more accurate predictions. This section focuses on the performance of different machine learning models trained on these datasets. The models are assessed for their ability to predict heat flux using metrics like Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), and R^2 scores to analyze the strengths and limitations of each dataset.

In the present investigation, machine learning was applied exclusively for heat flux prediction through regression analysis. The classification of boiling regimes was not handled using machine learning models; rather, regime transitions—such as those from incipient

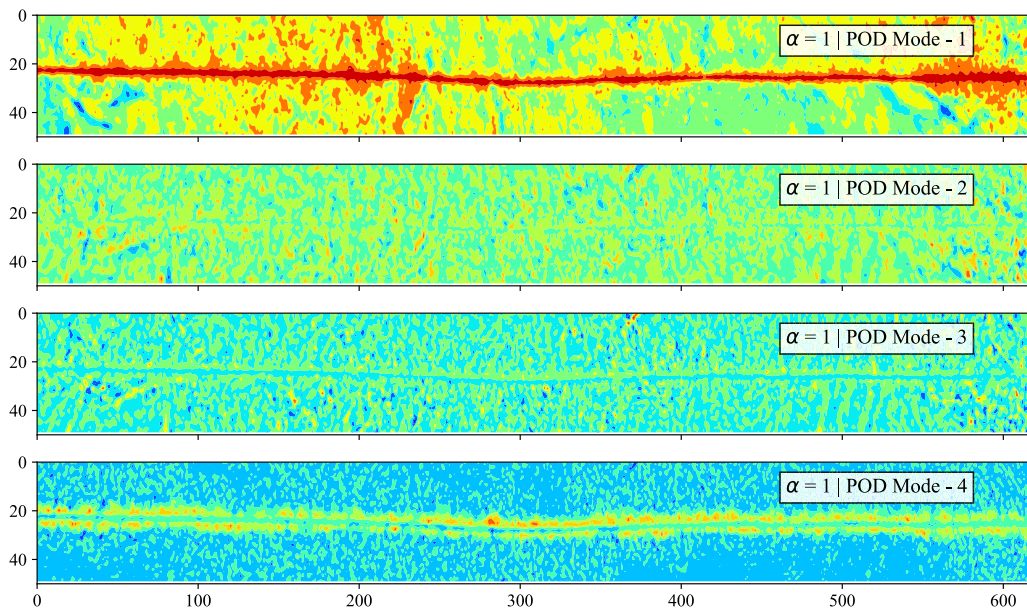


Fig. 16. POD Mode visualization.

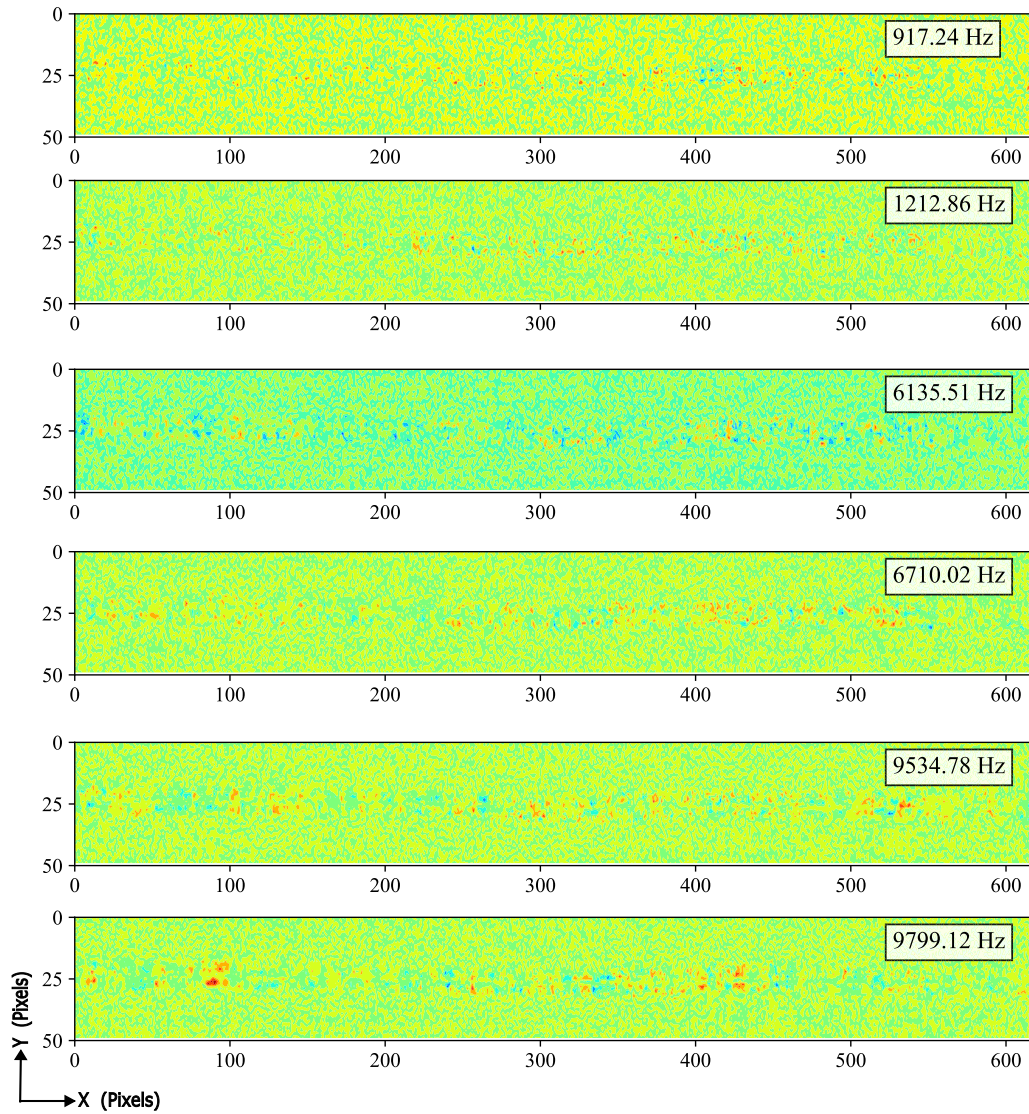


Fig. 17. Dynamic mode decomposition at various frequencies.

boiling to nucleate boiling and from nucleate boiling to critical heat flux (CHF) were identified through the frequency dynamics observed across the boiling curve. The heat flux ratio (α), defined as the ratio of heat flux to the CHF, was systematically varied from 0.06 to 1.0, capturing key stages of the boiling curve through 19 incremental steps. For each step, acoustic and vibration signals were recorded under steady-state conditions, resulting in approximately 160 recordings per step. This experimental procedure yielded a robust dataset consisting of 3093 recordings, effectively representing the boiling process.

Acoustic and vibration signals were simultaneously recorded at a sampling frequency of 150 kHz to capture dynamic bubble activity and surface vibrations. The raw acoustic and vibration data were preprocessed by subtracting the mean of each signal to normalize the data and improve feature clarity. The time-domain signals were converted into the frequency domain using the Welch method, producing 2049 frequency features for both the acoustic and vibration datasets. Dimensionality reduction through Principal Component Analysis (PCA) reduced the acoustic features to 3 principal components and the vibration features to 8 principal components while retaining 99% of the variance. This dimensionality reduction significantly improved computational efficiency while preserving essential information related to boiling dynamics.

For heat flux prediction, binary decision tree regression models were employed using a 5-fold cross-validation approach. The dataset

was split into 5 folds, with each fold serving as a validation set once, while the remaining 4 folds served as the training set. This cross-validation procedure improved the generalizability of the regression models, making them suitable for real-time heat flux prediction in boiling systems. The models demonstrated high prediction accuracy, with R^2 scores consistently exceeding 0.99, indicating that the models captured the variance in the dataset effectively. Acoustic data exhibited lower MSE, while vibration data showed lower MAPE, reflecting differences in their frequency characteristics and the nature of errors. These findings emphasize the robustness of the developed regression models in accurately predicting heat fluxes, even under varying boiling conditions.

3.6.1. Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is applied to reduce the dimensionality of the acoustic and vibration datasets, significantly lowering the complexity while retaining 99% of the original variance. The plot of cumulative explained variance ratio vs number of principal components for the acoustic and vibration data sets is depicted in Fig. 18. The cumulative explained variance ratio in the plot is a measure used in Principal Component Analysis (PCA) to quantify how much of the total variance in the original dataset is captured by the selected principal components. Each principal component explains a certain proportion of

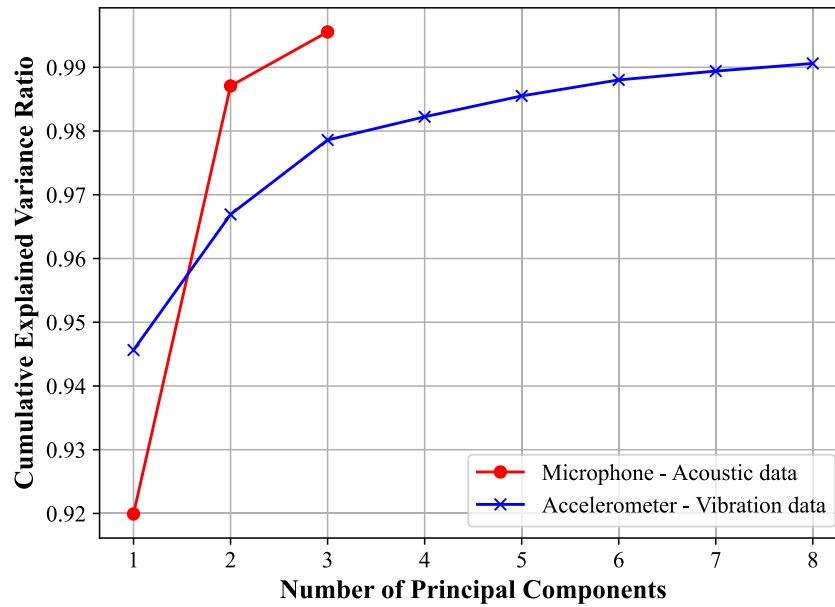


Fig. 18. Principal component analysis for acoustic and vibration data sets.

the variance in the data, and the cumulative explained variance ratio is the sum of these proportions up to the selected component.

For the acoustic dataset, consisting of 120 raw data features extracted from the frequency spectrum, the dimensionality is reduced to three principal components, resulting in a 97.5% reduction in dimensionality. In the case of the vibration dataset, which initially had 330 raw data features, PCA reduced the dimensionality to eight principal components, achieving a 97.6% reduction. These reductions are substantial, particularly when dealing with high-resolution spectral data, where many features may be correlated or carry redundant information.

This reduction enables the transformation of high-dimensional input signals into a low-dimensional representation that preserves the most significant underlying structure. PCA effectively filters out noise and less informative components, helping to focus machine learning models on the most relevant spectral patterns. By simplifying the dataset, PCA also reduces the risk of overfitting, enhances generalizability, and significantly lowers computational costs — which is critical for real-time or embedded boiling diagnostics systems where memory and processing power are constrained.

Importantly, PCA operates in a purely statistical domain and does not rely on assumptions regarding the physical properties of the system. It captures dominant variance through linear combinations of input features, making it particularly suitable for dynamic datasets like acoustic and vibration signals, where the physical origins of variance may be complex or unknown. The retained principal components correspond to the most influential spectral patterns in the dataset, ensuring that critical information about boiling dynamics and regime transitions is preserved.

Thus, PCA serves as a valuable preprocessing step, enabling efficient and accurate heat flux prediction from large-scale boiling datasets while maintaining interpretability and computational feasibility.

3.6.2. Performance of machine learning models for the boiling heat flux prediction

This paper includes a performance evaluation of simple binary decision tree models to predict heat flux under different boiling regimes. The decision tree models are utilized to assess their predictive accuracy, aligning with the performance evaluation metrics observed in the flow boiling heat flux prediction study by Barathula et al. [31]. The results of this study suggest that binary decision models are sufficient for

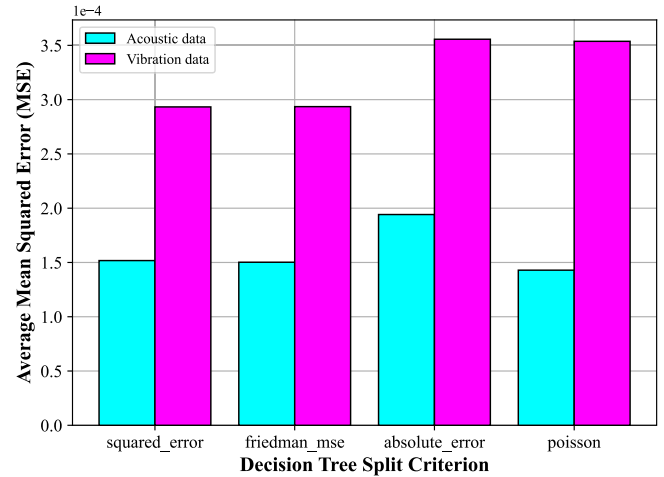


Fig. 19. Mean Squared Error (MSE) for acoustic and vibration data sets.

accurate heat flux prediction, as the complexity within the dataset is not prominent.

The performance comparison of decision tree split criteria (squared error, Friedman-mse, absolute error, and Poisson) on acoustic and vibration datasets provides a comprehensive view of the models' behavior across three metrics: Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), and R^2 score as shown in Figs. 19, 20 and 21. Across all cases, the R^2 scores for both datasets are consistently above 0.99, indicating that the models effectively capture the variance within the data, with only subtle differences emerging between the datasets. These high R^2 values suggest that despite some variations in the MSE and MAPE metrics, both the acoustic and vibration data are modeled accurately.

In terms of MSE, the vibration data consistently exhibits slightly higher error values across all split criteria when compared to the acoustic data, particularly with the absolute error and Poisson criteria. This could be attributed to the broader frequency range and the inherent complexity of the vibration data, leading to larger deviations in the predictions. However, given the minimal differences and the consistently high R^2 scores, these variations in MSE reflect only minor challenges in modeling the vibration data rather than significant difficulties.

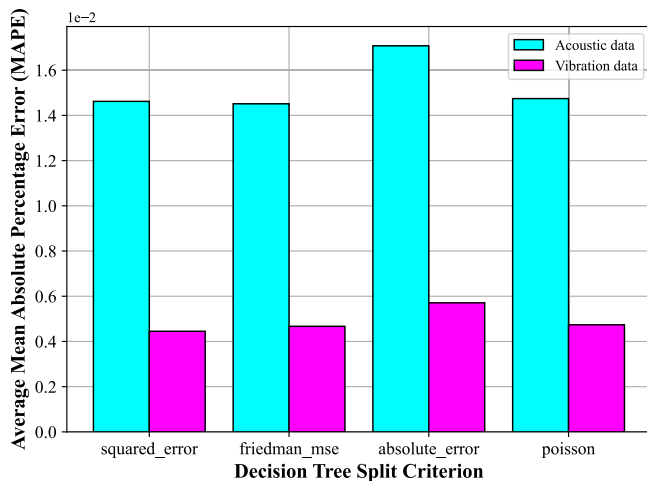


Fig. 20. Mean Absolute Percentage Error (MAPE) for acoustic and vibration data sets.

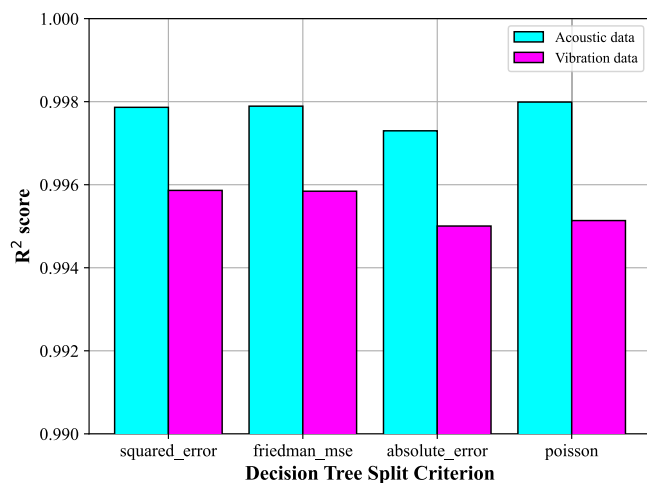


Fig. 21. R^2 Score for acoustic and vibration data sets.

The contradiction between MSE and MAPE sheds light on how the data is structured. The lower MSE for acoustic data suggests fewer large absolute deviations, whereas the higher MAPE indicates that relative errors (percentage-based) are more significant. This occurs because MAPE is sensitive to small actual values, meaning that even small absolute deviations in the acoustic data can lead to large percentage errors. On the other hand, the vibration data exhibits larger absolute errors (higher MSE), yet its relative percentage errors (MAPE) are smaller, likely due to larger values in the dataset that mitigate the impact of percentage-based errors.

Despite these differences, the overall performance remains strong across both datasets. The consistently high R^2 scores, combined with the modest variations in MSE and MAPE, suggest that both datasets are modeled effectively, with only subtle nuances in the way the errors manifest. The vibration data, while exhibiting slightly more complexity, is well-represented by the models, and the differences observed are relatively small.

4. Conclusions

This paper presents a comprehensive multi-scale approach for pool boiling regime classification, integrating acoustic, vibration, and imaging techniques. The study's findings offer significant contributions to understanding the boiling-induced phenomena across various heat flux levels.

Power spectral density analysis revealed that acoustic data showed a consistent rise in peak amplitudes with increasing heat flux, particularly highlighting enhanced bubble activity as CHF is approached. Vibration data exhibited a similar trend but indicated a plateau in peak amplitudes at higher heat flux levels, which coincided with CHF. The broader frequency range in the vibration spectra, compared to the acoustic spectra, resulted in more complex behavior, with notable energy concentration around 9 kHz, particularly before CHF is reached. The spectral power distribution analysis provided key insights, where the transition from nucleate boiling to film boiling could be identified through distinct shifts in spectral power from higher to lower frequencies.

POD revealed dominant spatial structures in the boiling phenomena. POD analysis demonstrated that as heat flux increased, boiling patterns evolved from isolated bubbles to more chaotic and widespread nucleation sites, ultimately leading to near-complete vapor coverage at CHF. DMD further supported these observations by identifying key frequencies associated with bubble dynamics and oscillations. DMD captured significant modes in the frequency range of 917.24 Hz to 9799.12 Hz, with peak activity at around 9534.78 Hz correlating with critical heat transfer points during boiling.

The performance evaluation of machine learning models confirmed that binary decision tree models are effective for heat flux prediction with relatively low error. Principal Component Analysis (PCA) reduced the acoustic dataset's dimensionality from 120 to 3 components and the vibration dataset from 330 to 8 components while retaining 99% of the variance in both cases. These results indicate a 97.5% reduction in the acoustic data complexity and a 97.6% reduction for the vibration data, making them computationally efficient for real-time monitoring. The decision tree models achieved high accuracy, with R^2 scores consistently exceeding 0.99 for both datasets, despite minor variations in Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE). The acoustic dataset demonstrated lower MSE values, while the vibration dataset showed lower MAPE values, suggesting differences in error sensitivity between absolute and relative scales.

In summary, this study demonstrates that the proposed multi-scale approach effectively captures the complex dynamics of pool boiling across different regimes and heat fluxes. Simple binary decision tree models are sufficient for predicting heat fluxes with low error, indicating that the complexity of the dataset is manageable. The combined use of acoustic, vibration, and imaging techniques, along with advanced decomposition methods, provides a robust framework for real-time classification and monitoring of boiling regimes, with important implications for improving the performance and safety of thermal management systems.

CRedit authorship contribution statement

Sreeram Barathula: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **R. Karthik:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Jaswanth K.K. Alapati:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **K. Srinivasan:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Teck Neng Wong:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

```

1 # ===== PCA and ML
2 # Section
3 # =====
4 # Step 1: Import necessary libraries
5 import numpy as np
6 from sklearn.decomposition import PCA
7 from sklearn.tree import
8   DecisionTreeRegressor
9 from sklearn.model_selection import KFold,
10   cross_val_score
11 from sklearn.metrics import
12   mean_squared_error,
13   mean_absolute_percentage_error,
14   r2_score, make_scorer
15
16 # Step 2: Apply PCA on both microphone (
17   acoustic) and accelerometer (vibration)
18   data
19 pca1 = PCA(n_components=0.99) # Retain 99%
20   variance for microphone data
21 pca1.fit(Mic.T)
22 Mic_PCA = pca1.components_.T
23
24 pca2 = PCA(n_components=0.99) # Retain 99%
25   variance for accelerometer data
26 pca2.fit(Acc.T)
27 Acc_PCA = pca2.components_.T
28
29 # Step 3: Evaluate cumulative explained
30   variance
31 cumulative_variance_ratio1 = np.cumsum(pca1
32   .explained_variance_ratio_)
33 cumulative_variance_ratio2 = np.cumsum(pca2
34   .explained_variance_ratio_)
35
36 # Step 4: Prepare target variable (e.g.,
37   heat flux or alpha value) and
38   initialize evaluation metrics
39 X = Mic_PCA # Microphone PCA components as
40   features
41 y = Target # Corresponding target
42   variable (e.g., heat flux)
43
44 # Custom MAPE scorer
45 mape_scorer = make_scorer(
46   mean_absolute_percentage_error,
47   greater_is_better=False)
48
49 # Step 5: 5-fold cross-validation setup
50 kf = KFold(n_splits=5, shuffle=True,
51   random_state=42)
52
53 # Step 6: Train decision tree models using
54   different criteria (squared_error,
55   absolute_error, etc.)
56 criteria = ['squared_error', 'friedman_mse',
57   'absolute_error', 'poisson']
58 results = {'MSE': {}, 'MAPE': {}, 'R2': {}}
59
60 for criterion in criteria:
61   tree_regressor = DecisionTreeRegressor(
62     criterion=criterion, random_state
63     =42)
64
65   # Cross-validation for Mean Squared
66   Error

```

```

41 mse_scores = -cross_val_score(
42   tree_regressor, X, y, cv=kf,
43   scoring='neg_mean_squared_error')
44 results['MSE'][criterion] = np.mean(
45   mse_scores)
46
47 # Cross-validation for Mean Absolute
48   Percentage Error (MAPE)
49 mape_scores = -cross_val_score(
50   tree_regressor, X, y, cv=kf,
51   scoring=mape_scorer)
52 results['MAPE'][criterion] = np.mean(
53   mape_scores)
54
55 # Cross-validation for R2 score
56 r2_scores = cross_val_score(
57   tree_regressor, X, y, cv=kf,
58   scoring='r2')
59 results['R2'][criterion] = np.mean(
60   r2_scores)
61
62 # Step 7: Train decision tree models on
63   accelerometer PCA data (Acc_PCA) using
64   the same procedure
65 X2 = Acc_PCA # Accelerometer PCA
66   components
67 results2 = {'MSE': {}, 'MAPE': {}, 'R2':
68   {}}
69
70 for criterion in criteria:
71   tree_regressor = DecisionTreeRegressor(
72     criterion=criterion, random_state
73     =42)
74
75   mse_scores = -cross_val_score(
76     tree_regressor, X2, y, cv=kf,
77     scoring='neg_mean_squared_error')
78   results2['MSE'][criterion] = np.mean(
79     mse_scores)
80
81   mape_scores = -cross_val_score(
82     tree_regressor, X2, y, cv=kf,
83     scoring=mape_scorer)
84   results2['MAPE'][criterion] = np.mean(
85     mape_scores)
86
87   r2_scores = cross_val_score(
88     tree_regressor, X2, y, cv=kf,
89     scoring='r2')
90   results2['R2'][criterion] = np.mean(
91     r2_scores)
92
93 # Step 8: Compare results using bar plots (
94   MSE, MAPE, and R2 scores for both
95   datasets)
96 # Criteria and results for comparison are
97   prepared for visualization
98
99 # ===== Code for POD
100   and DMD
101   =====
102
103 # Step 1: Import necessary libraries for
104   handling images, matrices, and plots
105 import tifffile as tiff
106 import numpy as np
107 import matplotlib.pyplot as plt
108 from scipy.linalg import svd, eig
109 from scipy.signal import welch

```

```

81 # Step 2: Initialize the list of alpha
82 values to process
83 Alpha_values = np.array([0.08, 0.11, ...,
84 1.0]) # (Choose specific alpha values)
85 # Step 3: Loop over each alpha value
86 for Alpha in Alpha_values:
87     Data = [] # Initialize an empty list
88     to store image data
89
90     # Step 3a: Load images corresponding to
91     the current alpha value
92     for i in range(number of images):
93         image_path = f"C:\\\\.....
94         .tif"
95         img = tiff.imread(image_path) #
96         Load the image
97         x, y, width, height = 180, 45, 620,
98         50 # Crop region parameters
99         cropped_img = np.array(img[y:y+
100         height, x:x+width]).ravel() #
101         Flatten cropped image
102         Data.append(cropped_img) # Store
103         in the list
104
105     # Step 3b: Create the image data matrix
106     (A)
107     Img_Data = np.vstack(Data).T # Stack
108     the image vectors into a matrix of
109     shape (pixels, time)
110
111 # ===== POD SECTION
112 =====
113
114 # Step 4: Compute the mean image and
115 visualize it
116 Img_Mean = np.mean(Img_Data, axis=1,
117 keepdims=True).reshape(height,
118 width) # Mean image
119 plt.figure(figsize=(width/height, 1.5))
120 plt.contourf(Img_Mean, cmap='jet')
121 plt.text(x=width * 0.9, y=height * 0.2,
122 s=f"$\\alpha$={Alpha}", fontsize
123 =14, color='black')
124 plt.tight_layout()
125 plt.show()
126
127 # Step 5: Perform Singular Value
128 Decomposition (SVD)
129 U, S, V = svd(Img_Data, full_matrices=
130 False) # Decompose A into U, S, V
131
132 # Step 6: Plot the singular values to
133 observe the energy content
134 plt.plot(range(len(S)), S, marker='o',
135 color='blue')
136 plt.xlim([0, 4])
137 plt.xlabel('Eigenvalue_Index', fontsize
138 =14, fontweight='bold')
139 plt.ylabel('Eigenvalue', fontsize=14,
140 fontweight='bold')
141 plt.grid()
142 plt.tight_layout()
143 plt.show()
144
145 # Step 7: Visualize the first few POD
146 modes using contour plots
147 num_modes = 4
148 fig, axes = plt.subplots(num_modes, 1,
149 figsize=(10, 4 * 1.5), sharex=True)

```

```

125 for n in range(num_modes):
126     ax = axes[n]
127     ax.contourf(U[:, n].reshape(50,
128 620), cmap='jet') # Reshape
129     and visualize the mode
130     ax.text(x=620 * 0.75, y=50 * 0.25,
131     s=f"$\\alpha$={Alpha}POD_Mode-{
132     n+1}", fontsize=14, color='
133     black')
134     ax.set_ylim([50, 0])
135     plt.tight_layout()
136     plt.show()
137
138 # Step 8: Perform frequency analysis
139 using Welch method on time
140 coefficients
141 Freq, FFT = welch(V[:, 0], fs=40000,
142 nperseg=2001, detrend='linear',
143 scaling='spectrum') # FFT on time
144 coefficients
145
146 # Step 9: Plot the FFT result
147 plt.plot(Freq, FFT/np.max(FFT)) #
148 Normalize and plot
149 plt.xlim([0, 12000])
150 plt.xlabel('Frequency(Hz)', fontsize
151 =14, fontweight='bold')
152 plt.ylabel('Amplitude$(V^2/Hz)$',
153 fontsize=14, fontweight='bold')
154 plt.ylim([0, None])
155 plt.show()
156
157 # ===== DMD SECTION
158 =====
159
160 # Step 10: Create time-lagged matrices
161 A1 and A2 for DMD
162 A1 = Img_Data[:, :-1] # All columns
163 except the last
164 A2 = Img_Data[:, 1:] # All columns
165 except the first
166
167 # Step 11: Perform SVD on A1
168 U_dmd, S_dmd, V_dmd = svd(A1,
169 full_matrices=False)
170
171 # Step 12: Reduce the matrices to the
172 desired number of modes
173 num_modes = 4 # Select the number of
174 modes to retain
175 U_dmd = U_dmd[:, :num_modes]
176 S_dmd = np.diag(S_dmd[:num_modes])
177 V_dmd = V_dmd[:num_modes, :]
178
179 # Step 13: Compute the reduced system
180 matrix L
181 L = U_dmd.T @ A2 @ V_dmd.T @ np.linalg.
182 inv(S_dmd)
183
184 # Step 14: Perform eigen decomposition
185 of L
186 eigvals, W = eig(L) # Eigenvalues (
187 frequencies) and eigenvectors
188
189 # Step 15: Compute the DMD modes
190 DMD_modes = A2 @ V_dmd.T @ np.linalg.
191 inv(S_dmd) @ W
192
193 # Step 16: Compute DMD frequencies from
194 eigenvalues

```

```

169     time_step = 1 / 40000 # Assuming the
        time step is related to the frame
        rate
170     frequencies = np.imag(np.log(eigvals))
        / (2 * np.pi * time_step)
171
172     # Step 17: Sort the modes based on the
        frequency magnitude
173     sorted_indices = np.argsort(np.abs(
        frequencies)) # Sort by magnitude
        of frequency
174     frequencies = frequencies[
        sorted_indices]
175     DMD_modes = DMD_modes[:, sorted_indices
        ]
176
177     # Step 18: Visualize the DMD modes
        corresponding to selected
        frequencies
178     fig, axes = plt.subplots(num_modes, 1,
        figsize=(10, 4 * 1.5), sharex=True)
179     for n in range(num_modes):
180         ax = axes[n]
181         ax.contourf(np.abs(DMD_modes[:, n].
        reshape(50, 620)), cmap='hot')
        # Reshape and visualize the
        mode
182         ax.text(x=620 * 0.75, y=50 * 0.25,
        s=f"Freq: {frequencies[n]:.2f} Hz, DMD Mode {n+1}",
        fontsize=14, color='black')
183         ax.set_ylim([50, 0])
184     plt.tight_layout()
185     plt.show()

```

Data availability

Data will be made available on request.

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